

Mock-up Test

This test is only for preparation of Computer-based Test (CBT). Solutions to these questions will be discussed in the lecture.

1. Consider the following database:

person (driver_ID, name, address, phone)
car (VIN, model, year, color)
owns (driver_ID, VIN)
accident (report_number, VIN, date, location)
participated (report_number, VIN, driver_ID, damage_amount)

Identify primary and foreign key in accident relation

- *accident (report_number PK, VIN FK, date, location)*
- *accident (report_number PK, VIN PK FK, date, location)*
- *accident (report_number FK, VIN PK FK, date, location)*
- *accident (report_number PK FK, VIN PK FK, date, location)*
- *All the other answers are incorrect.*

2. Consider the following database:

person (driver_ID, name, address, phone)
car (VIN, model, year, manufacturer)
owns (driver_ID, VIN)
accident (report_number, VIN, date, location)
participated (report_number, VIN, driver_ID, damage_amount)

Identify the multi-valued attributes.

- *Name, address, date*
- *Name, phone, manufacturer*
- *Name, address, phone*
- *Name, report_number, phone*

3. Consider the following database

INDIVIDUAL{BIRTH_DATE, FIRST_NAME, LAST_NAME, CUST_ID PK FK}
CUSTOMER {CUST_ID PK, ADDRESS, CITY, CUST_TYPE_CD, FED_ID, POSTAL_CODE, STATE}
ACCOUNT{ACCOUNT_ID PK, AVAIL_BALANCE, CLOSE_DATE, LAST_ACTIVITY_DATE,
OPEN_DATE, PENDING_BALANCE, STATUS, CUST_ID FK, OPEN_BRANCH_ID, OPEN_EMP_ID,
PRODUCT_CD}
BUSINESS{INCorp_DATE, NAME, STATE_ID, CUST_ID PK FK}

Identify the cardinality of the relationship from CUSTOMER to BUSINESS.

- *1:M.*
- *M:N.*
- *M:1.*
- *1:1*

4. Consider the following database

INDIVIDUAL{BIRTH_DATE, FIRST_NAME, LAST_NAME, CUST_ID PK FK}
CUSTOMER {CUST_ID PK, ADDRESS, CITY, CUST_TYPE_CD, FED_ID, POSTAL_CODE, STATE}
ACCOUNT{ACCOUNT_ID PK, AVAIL_BALANCE, CLOSE_DATE, LAST_ACTIVITY_DATE,
OPEN_DATE, PENDING_BALANCE, STATUS, CUST_ID FK, OPEN_BRANCH_ID, PRODUCT_CD}
EMPLOYEE{EMP_ID PK, END_DATE, FIRST_NAME, LAST_NAME, START_DATE, TITLE, DEPT_ID}

Identify the cardinality of the relationship between INDIVIDUAL and EMPLOYEE, including obligation.

- *All the other answers are incorrect.*
- *1:M, dashed line on both ends.*
- *1:M, dashed line towards person only.*
- *1:M, dashed line towards car only.*
- *1:M, solid line towards both ends.*

5. Consider a One-to-One relationship between the following entities: employee and Employee_ID_Card. Obligation between the two entities is represented with solid lines on both ends.

Which of the following statements is correct?

- *Each Employee_ID_Card belongs to at least one employee*
- *Each employee has at most one Employee_ID_Card.*
- *Each employee may have 1 or many Employee_ID_Card.*
- *Each Employee_ID_Card belongs exactly to one employee.*
- *All the other answers are incorrect*

6. Which of the following statements about SQL entity is correct?

- *Values Are Atomic i.e., columns in a relational table are not repeating group or arrays.*
- *Column Values Are of the Same Kind.*
- *Each Row is Unique.*
- *Each Column Has a Unique Name.*
- *All the statements are correct.*

7. Consider the following database

person {driver_ID, name, address, date_of_Birth, Phone_number}
car {VIN, model, year}
owns {driver_ID, VIN}
accident {report_number, VIN, date, location}
participated {report_number, VIN, driver_ID, damage_amount}

Determine the domain of all attributes in the entity "person".

- *int, numeric, text, datetime, numeric*
- *int, varchar, varchar, datetime, numeric.*
- *int, int, char, datetime, numeric.*
- *int, int, text, datetime, numeric*
- *All the other answers are incorrect*

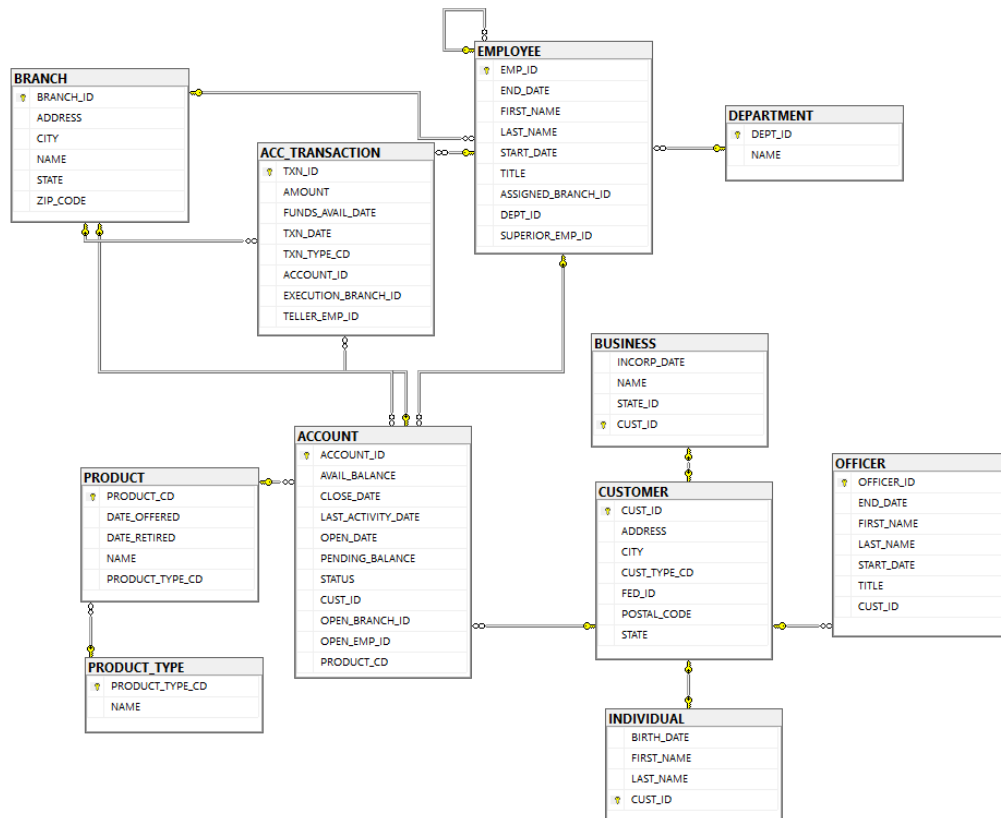
8. Consider the following database

INDIVIDUAL{BIRTH_DATE, FIRST_NAME, LAST_NAME, CUST_ID}
CUSTOMER{CUST_ID, ADDRESS, CITY, CUST_TYPE_CD, FED_ID, POSTAL_CODE, STATE}
ACCOUNT{ACCOUNT_ID PK, AVAIL_BALANCE, CLOSE_DATE, LAST_ACTIVITY_DATE, OPEN_DATE, PENDING_BALANCE, STATUS, CUST_ID FK, OPEN_BRANCH_ID, OPEN_EMP_ID, PRODUCT_CD}

Find the customer IDs of the customers whose available balance across all their accounts is greater or equal to the maximum available balance for all accounts in the database.

- *select CUST_ID, sum(AVAIL_BALANCE)
from ACCOUNT
group by CUST_ID
having sum(AVAIL_BALANCE) >= (
select max(AVAIL_BALANCE) from ACCOUNT)*
- *select CUST_ID, sum(AVAIL_BALANCE)
from ACCOUNT
group by CUST_ID
having sum(AVAIL_BALANCE) >= (
select min(AVAIL_BALANCE) from ACCOUNT)*
- *select CUST_ID, sum(AVAIL_BALANCE)
from ACCOUNT
group by CUST_ID
having sum(AVAIL_BALANCE) >= (
select avg(AVAIL_BALANCE) from ACCOUNT)*
- *none of the above query is correct*

9. Consider the following database and select the correct option for the given query.



```
SELECT I.FIRST_NAME, I.LAST_NAME
FROM CUSTOMER C INNER JOIN
INDIVIDUAL I ON
C.CUST_ID=I.CUST_ID
WHERE I.CUST_ID IN
(
SELECT CUST_ID
FROM ACCOUNT
WHERE AVAIL_BALANCE >=
(SELECT AVG(AVAIL_BALANCE) FROM ACCOUNT)
)
```

- Find the first and last names of all individuals whose available balance is greater or equal to the average available balance.
- Find the first and last names of all individuals whose available balance is the highest.
- Find the first and last names of all individuals whose available balance is less than the average available balance
- Find the first and last names of all individuals whose available balance is greater than the average available balance

10. Select **True/False** for the statement: A FOREIGN KEY is a field (or collection of fields) in one table, that refers to the PRIMARY KEY in another table.

- True
- False